

Number Power

Every graded set, every week — 4 weeks, 20 worksheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

How Big Is a Million

Read the place, then say what the digit is worth. Type digits only — no commas needed.

✦ Warm-Up 6 ITEMS

How many tens in 100?

How many hundreds in 1,000?

How many thousands in 10,000?

10×100

100×100

$1,000 \times 10$

◆ Core GRADED • SHOW YOUR WORKING

1. In 3,472,861 — what is the 4 worth? (It sits in the hundred-thousands place)

2. In 3,472,861 — what is the 7 worth? (Ten-thousands place)

3. In 5,208,043 — what is the 2 worth? (Count the places from the right)

4. How many thousands make 4,000,000? ($4,000 \times 1,000 = 4,000,000$)

5. Write six million forty thousand five (6,040,005 — hold the empty places with zeros)

★ Challenge NOT GRADED

C1. How many zeros in one million? (1,000,000)

C2. $1,000 \times 1,000$ (A thousand thousands)

Ten Times Bigger

Multiplying by ten shifts every digit one place left. Watch the zeros.

Warm-Up 6 ITEMS

40×10

700×10

60×100

$8 \times 1,000$

$3,200 \div 10$

$45,000 \div 1,000$

Core GRADED • SHOW YOUR WORKING

1. 230×100 (Two more zeros)

.....

2. $9,000 \div 100$ (Take two zeros away)

.....

3. $56 \times 1,000$ (Three more zeros)

.....

4. $740,000 \div 10,000$ (Four zeros off)

.....

5. How many times bigger is 800,000 than 800? (Three places left)

.....

Challenge NOT GRADED

C1. The 6 in 6,000 is worth how many times the 6 in 60? (Two places apart)

.....

C2. How many 10,000s make one million? ($1,000,000 \div 10,000$)

.....

Compare, Order, Round

Rounding depends on which place you were asked for. Read the question twice.

✦ Warm-Up 6 ITEMS

Round 47 to the nearest ten

Round 83 to the nearest ten

Round 65 to the nearest ten

Round 128 to the nearest ten

Round 254 to the nearest ten

Round 999 to the nearest ten

◆ Core GRADED • SHOW YOUR WORKING

1. Round 47,382 to the nearest thousand (382 down beats 618 up)

2. Round 47,382 to the nearest ten thousand (Now the ends are 40,000 and 50,000)

3. Round 863,209 to the nearest hundred thousand (63,209 up to 900,000 is the shorter trip)

4. Larger number: 408,916 or 480,169? Type it. (Compare the ten-thousands place)

5. Round 2,499 to the nearest hundred (Look at the tens digit only)

★ Challenge NOT GRADED

C1. Largest number that rounds to 5,000 at the nearest thousand (One less than the halfway point up)

C2. Smallest number that rounds to 5,000 at the nearest thousand (Exactly halfway rounds up)

Order of Operations

Build the groups first, then add and subtract what's left loose.

✦ Warm-Up 6 ITEMS

$3 + 4 \times 2$

$10 - 2 \times 3$

$12 \div 4 + 5$

$6 + 6 \div 2$

$2 \times 3 + 4 \times 5$

$20 - (4 + 6)$

◆ Core GRADED • SHOW YOUR WORKING

1. $5 + 3 \times 4$ (Multiply first: $3 \times 4 = 12$)

2. $(5 + 3) \times 4$ (Brackets overrule everything)

3. $18 \div (3 + 6)$ ($3 + 6 = 9$ first)

4. $40 - 6 \times 5 + 2$ ($40 - 30 + 2$)

5. $3 \times (12 - 4) \div 6$ ($3 \times 8 = 24$, then $\div 6$)

★ Challenge NOT GRADED

C1. $2 + 3 \times 4 - 6 \div 2$ ($2 + 12 - 3$)

C2. $(8 + 4) \times (9 - 6)$ (12×3)

Powers & Expression Duel

Enrichment day. The little raised number counts copies, not multiples.

★ Challenge NOT GRADED

C1. 2^3 (2 × 2 × 2)

C2. 5^2 (5 × 5)

C3. 10^4 (A 1 with four zeros)

C4. $3^2 + 4^2$ (9 + 16)

C5. $2^5 - 2^3$ (32 - 8)

C6. Bigger value: 4^3 or 3^4 ? Type it. (64 against 81)

C7. $10^6 \div 10^3$ (Six zeros take away three)

C8. How many zeros in 10^8 ? (The exponent counts them)

Week 1 Answers

Mission 03 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

1.1 • How Big Is a Million

OUT OF 11

Warm-Up • 10 | 10 | 10 | 1000 | 10000 | 10000

Core • 400000 | 70000 | 200000 | 4000 | 6040005

Challenge • 6 | 1000000

1.2 • Ten Times Bigger

OUT OF 11

Warm-Up • 400 | 7000 | 6000 | 8000 | 320 | 45

Core • 23000 | 90 | 56000 | 74 | 1000

Challenge • 100 | 100

1.3 • Compare, Order, Round

OUT OF 11

Warm-Up • 50 | 80 | 70 | 130 | 250 | 1000

Core • 47000 | 50000 | 900000 | 480169 | 2500

Challenge • 5499 | 4500

1.4 • Order of Operations

OUT OF 11

Warm-Up • 11 | 4 | 8 | 9 | 26 | 10

Core • 17 | 32 | 2 | 12 | 4

Challenge • 11 | 36

Fri • Powers & Expression Duel

ENRICHMENT

Challenge • 8 | 25 | 10000 | 25 | 24 | 81 | 1000 | 8

Rounding to Any Place

Find the place, look one to its right, decide.

 Warm-Up 6 ITEMS

Round 48 to the nearest ten

Round 43 to the nearest ten

Round 250 to the nearest hundred

Round 149 to the nearest hundred

Round 75 to the nearest ten

Round 612 to the nearest ten

 Core GRADED • SHOW YOUR WORKING

1. Round 4,829 to the nearest hundred

2. Round 4,829 to the nearest thousand

3. Round 27,500 to the nearest thousand

4. Round 96,412 to the nearest ten thousand

5. Round 8,950 to the nearest hundred

 Challenge NOT GRADED

C1. A number rounds to 300 to the nearest hundred. Its smallest possible value

C2. And its largest possible whole value

Estimate Before You Compute

Round first. Then the real answer has something to be checked against.

Warm-Up 6 ITEMS

Estimate $48 + 31$ by rounding both to tens

Estimate $62 - 19$

Estimate 19×21

Estimate $81 \div 9$

Estimate $297 + 104$

Estimate $52 + 48$

Core GRADED • SHOW YOUR WORKING

1. Estimate $4,812 + 3,190$ to the nearest thousand

.....

2. Estimate $612 - 388$ to the nearest hundred

.....

3. Estimate 187×4

.....

4. True value of 187×4

.....

5. Estimate $3,964 \div 4$

.....

Challenge NOT GRADED

C1. Estimate the cost of 19 items at \$4.95 each

.....

C2. Someone estimates 512×6 as 300. Type the real answer

.....

How Wrong Is the Estimate

An estimate you never compare to anything teaches you nothing.

✦ Warm-Up 6 ITEMS

Estimate of $48 + 31$

True value of $48 + 31$

The difference

True value of 19×21

Estimate of 19×21

The difference

◆ Core GRADED • SHOW YOUR WORKING

1. True value of $612 - 388$

.....

2. Estimate of $612 - 388$ to the nearest hundred

.....

3. How far off was that estimate

.....

4. True value of $4,812 + 3,190$

.....

5. How far off was the thousand estimate

.....

★ Challenge NOT GRADED

C1. Rounding both numbers up always makes the estimate too high or too low — type high or low

.....

C2. Estimate 39×41 , then give the true value

.....

Big Numbers in Context

Millions stop being abstract when you attach them to something real.

 Warm-Up 6 ITEMS

Zeros in one thousand _____	Zeros in one million _____	Thousands in a million _____
Hundreds in a thousand _____	Tens in a thousand _____	Zeros in ten thousand _____

 Core GRADED • SHOW YOUR WORKING

1. Days in about a million minutes, rounded to the nearest hundred (A million minutes is about 694 days)

2. Dots per page 1,000 — pages for a million dots

3. Seconds in an hour

4. Hours in a week

5. Minutes in a day

 Challenge NOT GRADED

C1. About how many years is a million days, to the nearest thousand
.....

C2. A stack of 100 sheets is 1 cm. Height in metres of a million sheets
.....

A Million Dots, Measured

Prove the estimate. Count one square, scale it up, defend the number.

 Warm-Up 2 ITEMS

Dots in a 10 by 10 square

Dots in ten such squares

 Core GRADED • SHOW YOUR WORKING

1. Squares of 100 dots needed for a million

.....

2. A 100 by 100 grid holds how many dots

.....

 Challenge NOT GRADED

C1. How many 100 by 100 grids make a million

.....

C2. If one grid is 20 cm wide, the width in metres of 100 side by side

.....

C3. Dots in a 1,000 by 1,000 grid

.....

Week 2 Answers

Mission 03 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

2.1 • Rounding to Any Place

OUT OF 11

Warm-Up • 50 | 40 | 300 | 100 | 80 | 610

Core • 4800 | 5000 | 28000 | 100000 | 9000

Challenge • 250 | 349

2.2 • Estimate Before You Compute

OUT OF 11

Warm-Up • 80 | 40 | 400 | 9 | 400 | 100

Core • 8000 | 200 | 800 | 748 | 1000

Challenge • 100 | 3072

2.3 • How Wrong Is the Estimate

OUT OF 11

Warm-Up • 80 | 79 | 1 | 399 | 400 | 1

Core • 224 | 200 | 24 | 8002 | 2

Challenge • high | 1599

2.4 • Big Numbers in Context

OUT OF 11

Warm-Up • 3 | 6 | 1000 | 10 | 100 | 4

Core • 700 | 1000 | 3600 | 168 | 1440

Challenge • 3000 | 100

Fri • A Million Dots, Measured

OUT OF 4

Warm-Up • 100 | 1000

Core • 10000 | 10000

Challenge • 100 | 20 | 1000000

Why Order Matters

Without an agreed order, one expression has several answers.

✦ Warm-Up 6 ITEMS

$2 + 3 \times 4$

$(2 + 3) \times 4$

$10 - 2 \times 3$

$(10 - 2) \times 3$

$6 + 6 \div 2$

$(6 + 6) \div 2$

◆ Core GRADED • SHOW YOUR WORKING

1. $5 + 4 \times 3 - 2$

2. $(5 + 4) \times (3 - 2)$

3. $20 - 3 \times 4 + 2$

4. $36 \div (2 + 4)$

5. $36 \div 2 + 4$

★ Challenge NOT GRADED

C1. $2 \times (3 + 4) \times 5$

.....

C2. Place brackets in $8 + 2 \times 5$ to make 50 — type the answer you get

.....

Brackets First

Whatever is inside the brackets happens before anything else.

Warm-Up 6 ITEMS

$$(4 + 5) \times 2$$

$$(9 - 3) \div 2$$

$$3 \times (2 + 2)$$

$$(7 + 3) \times 10$$

$$(12 - 4) \div 4$$

$$5 \times (10 - 6)$$

Core GRADED • SHOW YOUR WORKING

1. $(15 + 5) \div (2 + 2)$

.....

2. $4 \times (6 + 3) - 10$

.....

3. $100 - (25 \times 3)$

.....

4. $(8 \times 3) - (4 \times 2)$

.....

5. $2 \times (50 - 15)$

.....

Challenge NOT GRADED

C1. $((4 + 2) \times 3) - 8$

.....

C2. $120 \div (2 \times (3 + 3))$

.....

Multi-Step Expressions

Multiply and divide before you add and subtract. Left to right within each.

✦ Warm-Up 6 ITEMS

$12 \div 3 + 2$

$12 + 3 \times 2$

$20 \div 4 \times 2$

$18 - 6 \div 3$

$7 \times 2 + 6$

$30 \div 5 - 2$

◆ Core GRADED • SHOW YOUR WORKING

1. $48 \div 6 + 3 \times 4$

.....

2. $100 - 4 \times 12$

.....

3. $7 \times 8 - 6 \times 5$

.....

4. $64 \div 8 \div 2$

.....

5. $9 + 36 \div 4 - 3$

.....

★ Challenge NOT GRADED

C1. $144 \div 12 + 8 \times 3 - 5$

.....

C2. $2 \times 3 + 4 \times 5 + 6 \times 7$

.....

Write the Expression

Turn the sentence into symbols, then work it out.

✦ Warm-Up 6 ITEMS

Five more than three lots of four

Double seven, then add one

Ten less than six times three

Half of twenty, plus four

Three lots of nine

Twenty shared by four

◆ Core GRADED • SHOW YOUR WORKING

1. The sum of 8 and 4, multiplied by 3

.....

2. 8 added to 4 lots of 3

.....

3. 100 less the product of 7 and 9

.....

4. The difference of 20 and 8, divided by 4

.....

5. Six lots of the sum of 2 and 3

.....

★ Challenge NOT GRADED

C1. Three times the difference between 15 and 7

.....

C2. The sum of 12 and 8, divided by the difference of 9 and 4

.....

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

✦ Warm-Up 2 ITEMS

Round 4,829 to the nearest hundred

$2 + 3 \times 4$

◆ Core GRADED • SHOW YOUR WORKING

1. $(5 + 4) \times (3 - 2)$

.....

2. $48 \div 6 + 3 \times 4$

.....

3. Estimate 187×4

.....

4. Zeros in one million

.....

5. The sum of 8 and 4, multiplied by 3

.....

★ Challenge NOT GRADED

C1. $144 \div 12 + 8 \times 3 - 5$

.....

Week 3 Answers

Mission 03 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

3.1 • Why Order Matters

OUT OF 11

Warm-Up • 14 | 20 | 4 | 24 | 9 | 6

Core • 15 | 9 | 10 | 6 | 22

Challenge • 70 | 50

3.2 • Brackets First

OUT OF 11

Warm-Up • 18 | 3 | 12 | 100 | 2 | 20

Core • 5 | 26 | 25 | 16 | 70

Challenge • 10 | 10

3.3 • Multi-Step Expressions

OUT OF 11

Warm-Up • 6 | 18 | 10 | 16 | 20 | 4

Core • 20 | 52 | 26 | 4 | 15

Challenge • 31 | 68

3.4 • Write the Expression

OUT OF 11

Warm-Up • 17 | 15 | 8 | 14 | 27 | 5

Core • 36 | 20 | 37 | 3 | 30

Challenge • 24 | 4

Fri • Mid-Unit Quiz

OUT OF 7

Warm-Up • 4800 | 14

Core • 9 | 20 | 800 | 6 | 36

Challenge • 31

The Little Raised Number

An exponent counts how many times the base is multiplied by itself.

✦ Warm-Up 6 ITEMS

 2^2 3^2 5^2 2^3 10^2 4^2

◆ Core GRADED • SHOW YOUR WORKING

1. 3^3

2. 2^4

3. 5^3

4. 10^3

5. 2^5

★ Challenge NOT GRADED

C1. 2^8

C2. Which is bigger, 3^4 or 4^3 — type the value of the bigger

Powers of Ten

The exponent is the number of zeros. That is the whole shortcut.

Warm-Up 6 ITEMS

 10^1 10^2 10^3 10^4 Zeros in 10^5 10^6

Core GRADED • SHOW YOUR WORKING

1. 3×10^2

2. 7×10^3

3. 4×10^4

4. Write 5,000 as a digit times a power of ten — type the exponent

5. 25×10^2

Challenge NOT GRADED

C1. $10^3 \times 10^2$

C2. A million written as a power of ten — type the exponent

Expanded Form with Powers

Every place value is a power of ten in disguise.

Warm-Up 6 ITEMS

The value of the 4 in 400

The value of the 7 in 70

The value of the 3 in 3,000

$200 + 30 + 5$

$4,000 + 200 + 10$

The value of the 9 in 900

Core GRADED • SHOW YOUR WORKING

1. $6 \times 10^3 + 4 \times 10^2 + 2$

2. $3 \times 10^4 + 5 \times 10^2$

3. The value of the 8 in 84,000

4. $9 \times 10^2 + 9 \times 10^1 + 9$

5. 7×10^5

Challenge NOT GRADED

C1. $2 \times 10^5 + 6 \times 10^3 + 4 \times 10^1$

C2. The largest power of ten inside 47,300 — type its exponent

Error Journal Sweep

Mixed review of the whole mission. Fix only what repeats.

✦ Warm-Up 6 ITEMS

Round 149 to the nearest hundred

$2 + 3 \times 4$

10^3

Zeros in one million

5^2

$(2 + 3) \times 4$

◆ Core GRADED • SHOW YOUR WORKING

1. Round 96,412 to the nearest ten thousand

2. $48 \div 6 + 3 \times 4$

3. 4×10^4

4. Estimate $3,964 \div 4$

5. $6 \times 10^3 + 4 \times 10^2 + 2$

★ Challenge NOT GRADED

C1. $144 \div 12 + 8 \times 3 - 5$

C2. 2^8

Mission 03 Test

Twelve items plus the Big Question, answered out loud.

✦ Warm-Up 2 ITEMS

Round 612 to the nearest ten

3^2

◆ Core GRADED • SHOW YOUR WORKING

1. Round 4,829 to the nearest thousand

.....

2. $10 - 2 \times 3$

.....

3. $(15 + 5) \div (2 + 2)$

.....

4. $100 - 4 \times 12$

.....

5. 7×10^3

.....

6. 2^5

.....

7. Estimate $4,812 + 3,190$ to the nearest thousand

.....

8. The sum of 12 and 8, divided by the difference of 9 and 4

.....

★ Challenge NOT GRADED

C1. $2 \times 10^5 + 6 \times 10^3 + 4 \times 10^1$

C2. $2 \times 3 + 4 \times 5 + 6 \times 7$

Week 4 Answers

Mission 03 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

4.1 • The Little Raised Number

OUT OF 11

Warm-Up • 4 | 9 | 25 | 8 | 100 | 16

Core • 27 | 16 | 125 | 1000 | 32

Challenge • 256 | 81

4.2 • Powers of Ten

OUT OF 11

Warm-Up • 10 | 100 | 1000 | 10000 | 5 | 1000000

Core • 300 | 7000 | 40000 | 3 | 2500

Challenge • 100000 | 6

4.3 • Expanded Form with Powers

OUT OF 11

Warm-Up • 400 | 70 | 3000 | 235 | 4210 | 900

Core • 6402 | 30500 | 80000 | 999 | 700000

Challenge • 206040 | 4

Thu • Error Journal Sweep

OUT OF 11

Warm-Up • 100 | 14 | 1000 | 6 | 25 | 20

Core • 100000 | 20 | 40000 | 1000 | 6402

Challenge • 31 | 256

Fri • Mission 03 Test

OUT OF 10

Warm-Up • 610 | 9

Core • 5000 | 4 | 5 | 52 | 7000 | 32 | 8000 | 4

Challenge • 206040 | 68